

Full-Cohort Revised Thesis

MIMIC-IV v3.1 treatment-feasibility modelling after inclusion/exclusion criteria

Abstract

This revised thesis summary reports a full-cohort MIMIC-IV v3.1 modelling study after applying the study inclusion and exclusion criteria. The final modelling dataset contains 32,399 adult first ICU stays, not the previous 2,000-stay benchmark subset. The strongest rerun model was Stacked Classical with AUROC 0.755.

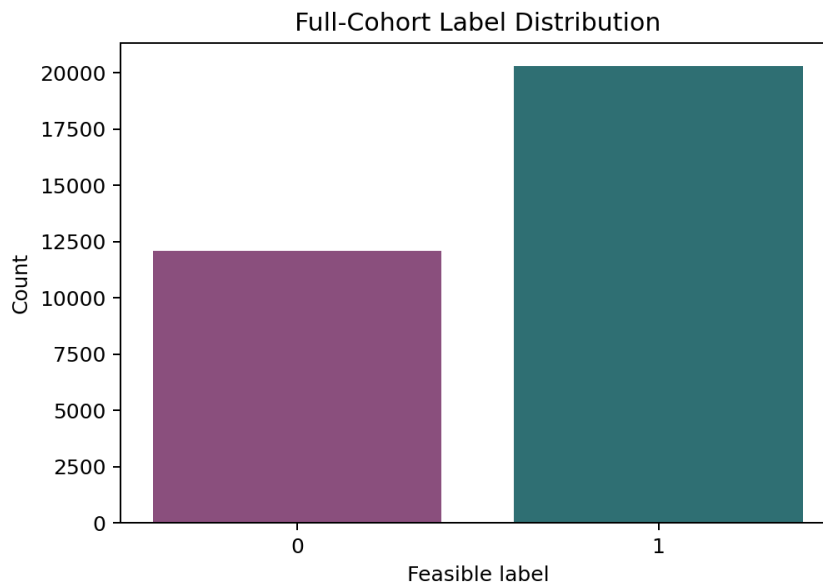
Methods

The cohort included adult first ICU stays with ICU length of stay at least 4 hours, non-missing discharge location, and an available primary ICD-10 diagnosis. First-24-hour laboratory and vital-sign features were extracted from the full eligible cohort. Missing feature values were quantified before median imputation and tokenization.

Quantity	Value
Eligible adult first ICU stays	32,399
Final modelling rows	32,399
Unique ICU stays	32,399
Unique subjects	32,399
Token tensor shape	(32399, 25, 4)
All eligible data used	yes

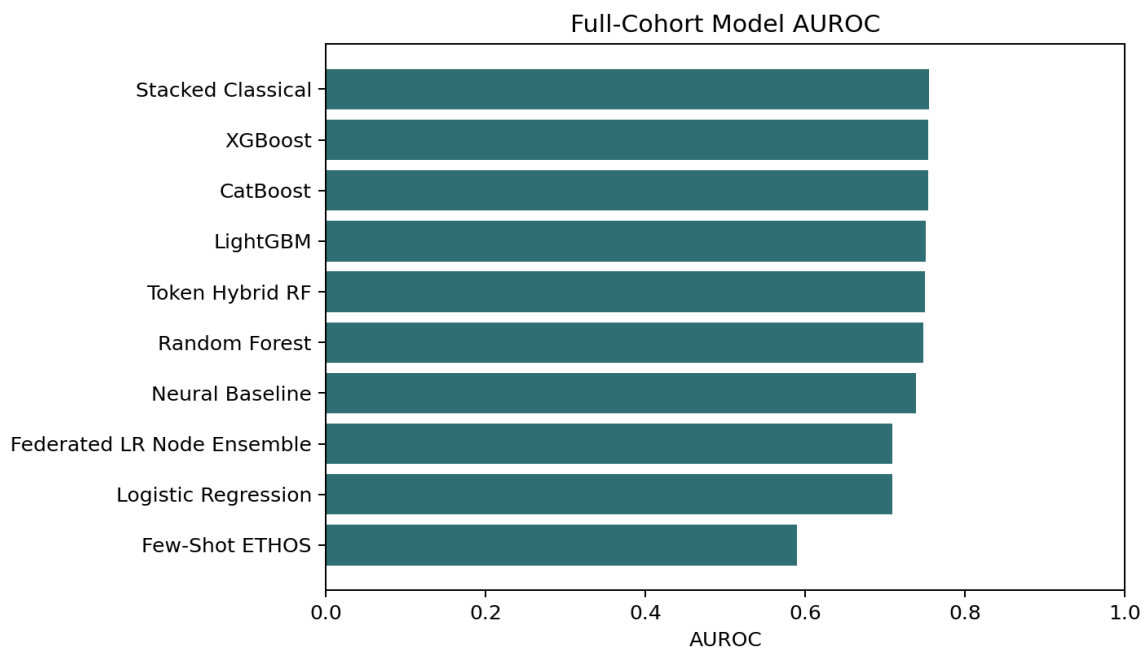
Results

Label distribution was 0: 12102, 1: 20297. Mean pre-imputation feature missingness was 25.2%; post-imputation missing cells in the modelling matrix were 0.



Model	Train/Val/Test	AUROC	AUPRC	Accuracy	Recall	Specificity	F1	Brier	ECE
Stacked Classical	20735/5184/6480	0.755	0.825	0.699	0.769	0.583	0.677	0.202	0.109
XGBoost	20735/5184/6480	0.753	0.825	0.697	0.761	0.589	0.675	0.189	0.020
CatBoost	20735/5184/6480	0.753	0.823	0.703	0.790	0.558	0.677	0.200	0.100

Model	Train/Val/Test	AUROC	AUPRC	Accuracy	Recall	Specificity	F1	Brier	ECE
LightGBM	20735/5184/6480	0.751	0.823	0.690	0.730	0.623	0.674	0.199	0.090
Token Hybrid RF	20735/5184/6480	0.749	0.820	0.698	0.772	0.573	0.674	0.192	0.031
Random Forest	20735/5184/6480	0.748	0.820	0.689	0.742	0.601	0.670	0.192	0.028
Neural Baseline	20735/5184/6480	0.739	0.814	0.675	0.682	0.663	0.664	0.194	0.015
Federated LR Node Ensemble	20735/5184/6480	0.709	0.778	0.678	0.746	0.565	0.656	0.215	0.109
Logistic Regression	20735/5184/6480	0.709	0.778	0.680	0.754	0.556	0.656	0.215	0.109
Few-Shot ETHOS	20735/5184/6480	0.590	0.668	0.590	0.642	0.504	0.570	0.250	0.127



Historical 2,000-stay benchmark comparisons are reported below only as context. They are not reused as full-cohort results.

Model	Full AUROC	Benchmark AUROC	Delta	Benchmark source
Stacked Classical	0.755	0.746	0.009	2,000-stay stacked BEST
XGBoost	0.753	0.694	0.059	2,000-stay Exp1
CatBoost	0.753	NA	NA	No directly comparable old benchmark metric
LightGBM	0.751	NA	NA	No directly comparable old benchmark metric
Token Hybrid RF	0.749	0.717	0.032	2,000-stay ETHOS-RF hybrid
Random Forest	0.748	0.762	-0.015	2,000-stay Exp1
Neural Baseline	0.739	0.652	0.087	2,000-stay Exp1
Federated LR Node Ensemble	0.709	0.547	0.162	2,000-stay federated experiment
Logistic Regression	0.709	0.724	-0.015	2,000-stay Exp1
Few-Shot ETHOS	0.590	0.611	-0.022	2,000-stay Exp1

Discussion

The revised framing is a full-cohort modelling study. The old 2,000-stay benchmark is not used as evidence for the main full-cohort claims and should only be cited as historical benchmark context.

Limitations

The feasibility outcome remains a proxy derived from discharge disposition, survival flag, and ICU length of stay. Feature extraction uses selected first-24-hour labs and vitals, and unmeasured features are imputed after missingness is recorded. Model results are single-split reruns unless otherwise stated.